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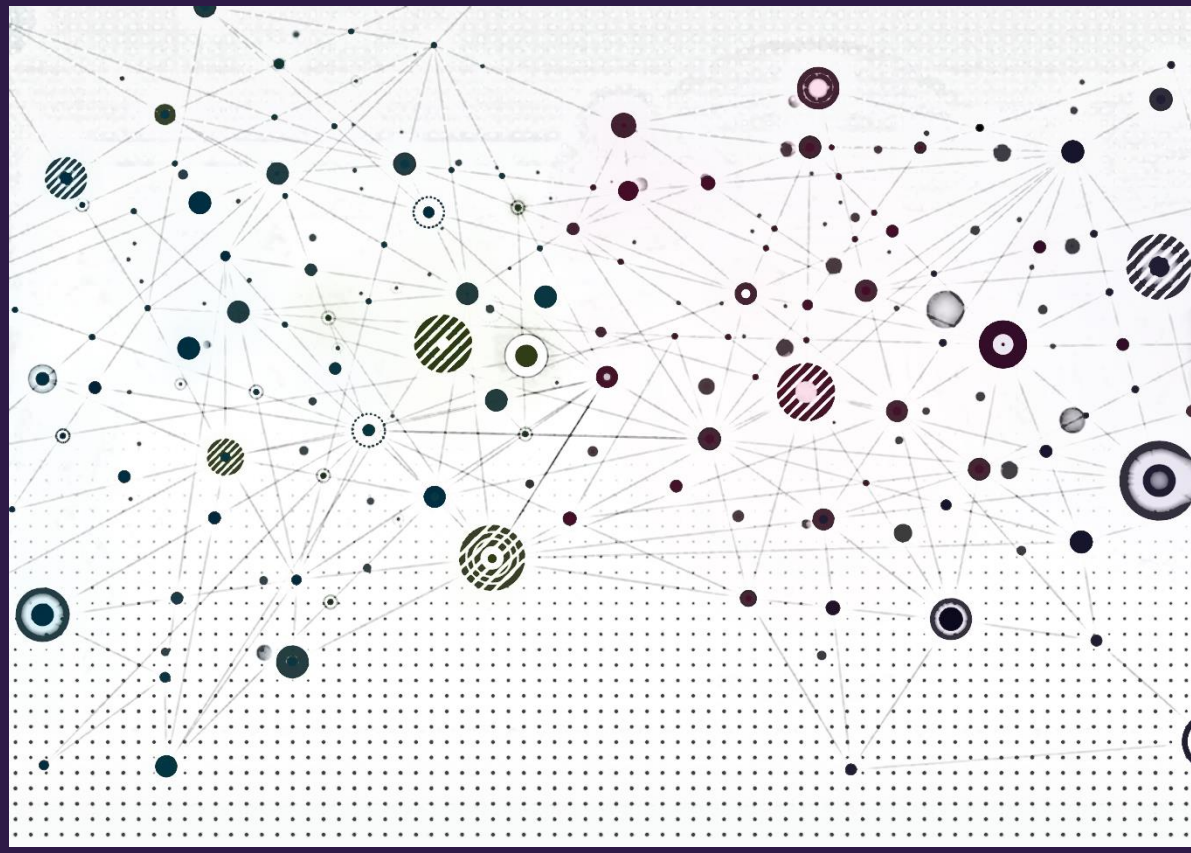
GNNs in practice



Overview

- Recap
- GNN prediction heads
- GNN training
- GNN performance evaluation
- Splitting our dataset

Mid-course survey

- Goal: adjust course pace & material to improve learning outcomes
- 3 multiple choice questions
- Anonymous
- https://clemson.instructure.com/courses/271779/quizzes/599347

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Course Project Review

READ ME!

Complete details on course project requirements are on Canvas:

ML4G-8810-Course-Project-Details.pdf

Course project

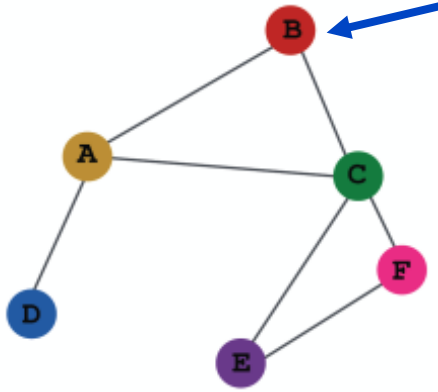
- Goals
  - Gain experience with implementing and evaluating graph machine learning models on a project that is meaningful to you
  - Create a lasting resource for your technical profile - lab PIs and hiring managers love these
- Three types of projects
  1. Real-world application of graph neural networks (GNNs)
  2. In-depth web-based tutorial on PyTorch Geometric (PyG) functionality
  3. Implementation of new research in PyG
- (Opt In) If you write a blog post for your project, I will add it to the course blog.

Recap

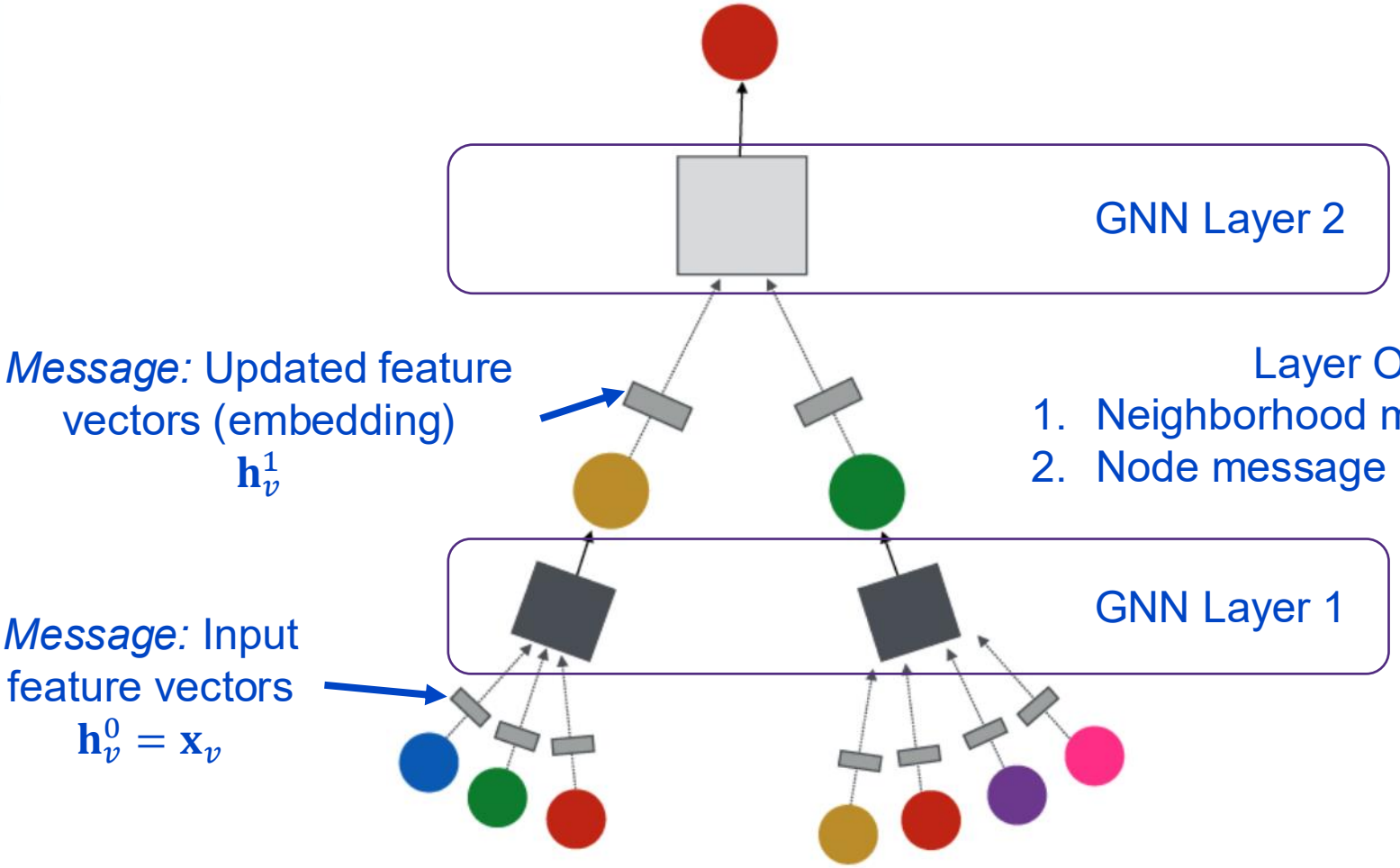
Message passing in GNNs

GNN Layer ≡ Aggregation → Update

- Different architectures under this perspective
- GCN, GraphSage, GAT, …



Input graph



Message: Updated feature vectors (embedding)
$\mathbf{h}_v^1$

Message: Input feature vectors
$\mathbf{h}_v^0 = \mathbf{x}_v$

Layer Operations
1. Neighborhood message aggregation
2. Node message (embedding) update

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Message aggregation and update as a GNN layer

- Combine the neighborhood message aggregation and node embedding update to form a single GNN layer
  1. Aggregation: each node aggregates messages (embeddings) of neighbor nodes

$$\mathbf{m}_{N(v)}^{(k)} = Agg^{(k)}\left(\left\{\mathbf{h}_u^{(k)}, \forall u \in N(v)\right\}\right)$$

  2. Update: each node updates its embedding (message for next layer)

$$\mathbf{h}_v^{(k+1)} = Update^{(k)}\left(\mathbf{h}_v^{(k)}, \mathbf{m}_{N(v)}^{(k)}\right)$$

  - The update usually includes a nonlinearity (activation), $\sigma(\cdot)$, such as ReLU, sigmoid

Graph convolutional networks

- A GCN can be written as

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)} \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}\right)$$

Recall: Graph augmented with self loops

- How to write this as aggregation → update?

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}$$

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)} \mathbf{m}_{N(v)}^{(k)}\right)$$

GraphSAGE (SAmple and aggreGatE) sampling

- In expectation (on average), we get embeddings similar to aggregating over all neighbors
  - Greatly reduces computational cost

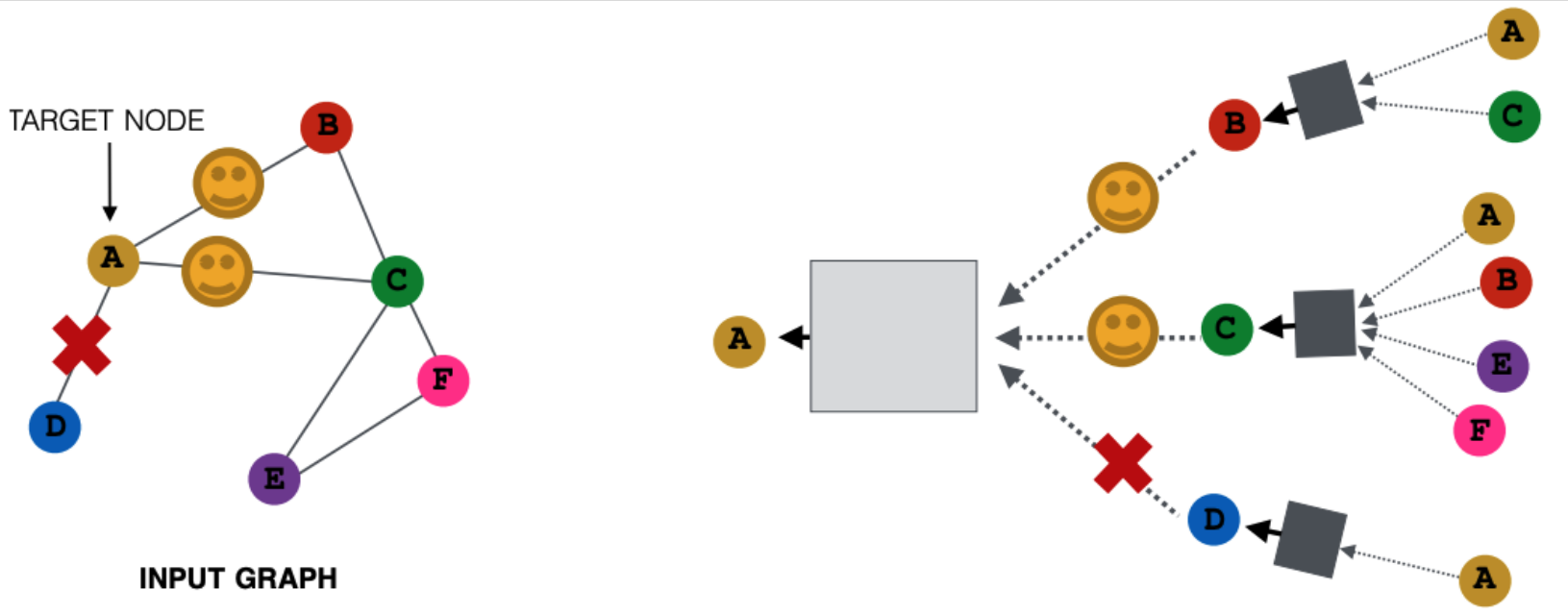


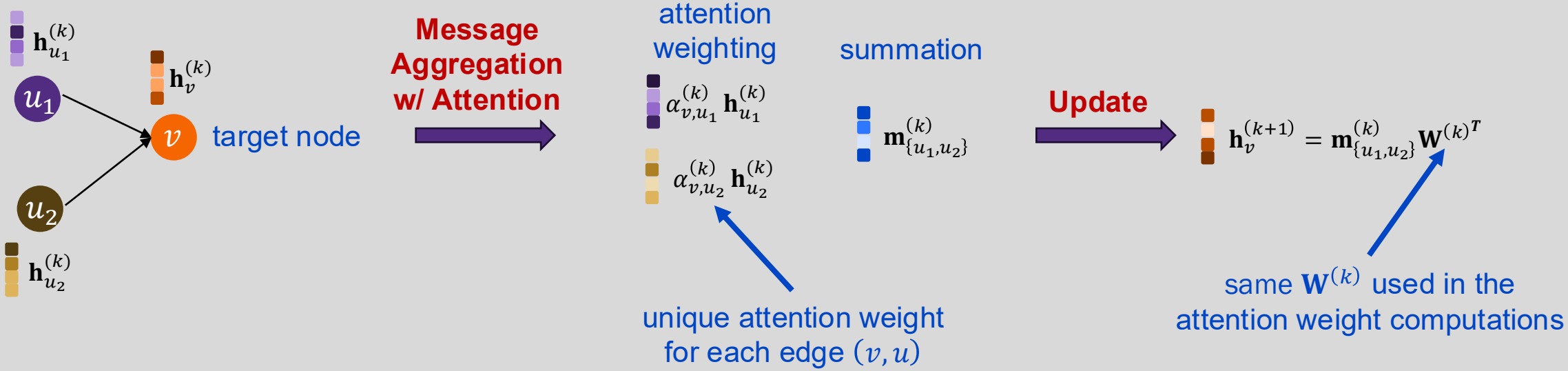
Image credit: Jure Leskovec, http://cs224w.stanford.edu

Graph attention with convolution update

- Formulate attention in the message passing framework

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k)} \mathbf{h}_u^{(k)}$$

$$\mathbf{h}^{(k+1)} = \mathbf{m}_{N(v)}^{(k)} \mathbf{W}^{(k)T}$$

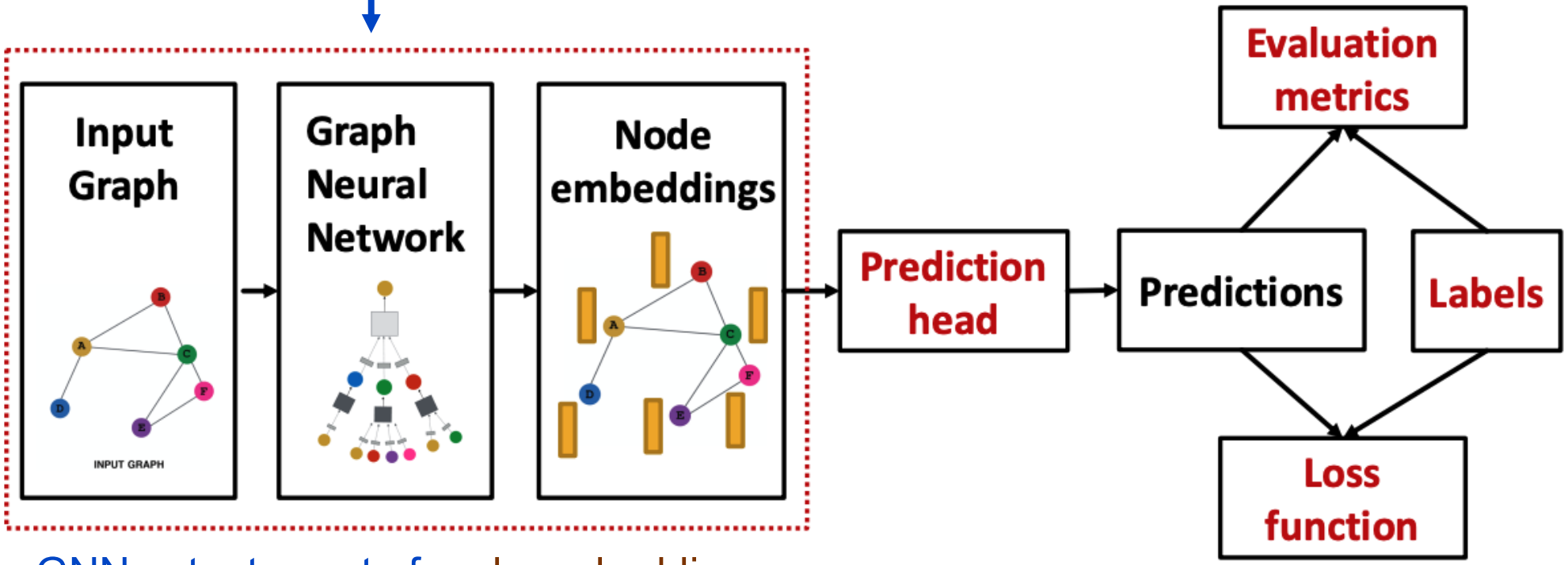


Why manipulate graphs

- We've assumed that the network graph = the GNN structure
- We might want to break this assumption if:
  - The input graph lacks features → apply feature augmentation (later lecture)
  - The graph is too sparse → add virtual nodes
  - The graph is too dense → apply GraphSAGE or other neighbor sampling
  - The graph is too large → sample subgraphs to compute embeddings (later lecture, time permitting)

GNN prediction heads

We've learned about these



GNN output: a set of node embeddings

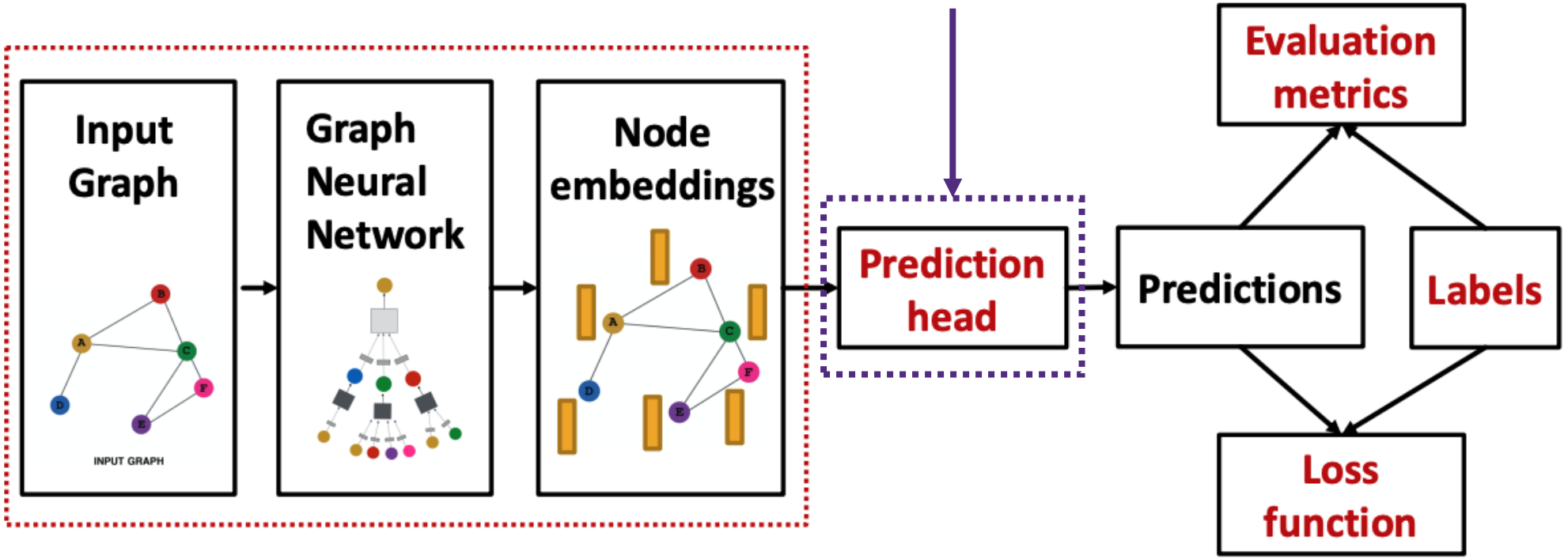
$$\left\{\mathbf{h}_v^{(K)} \; \forall v \in V\right\}$$

What do we do with these?

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Add “prediction heads”
(additional network layers) for:

- node-level tasks
- edge-level tasks
- graph-level tasks



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GNN prediction heads

Different tasks levels require different types of prediction heads

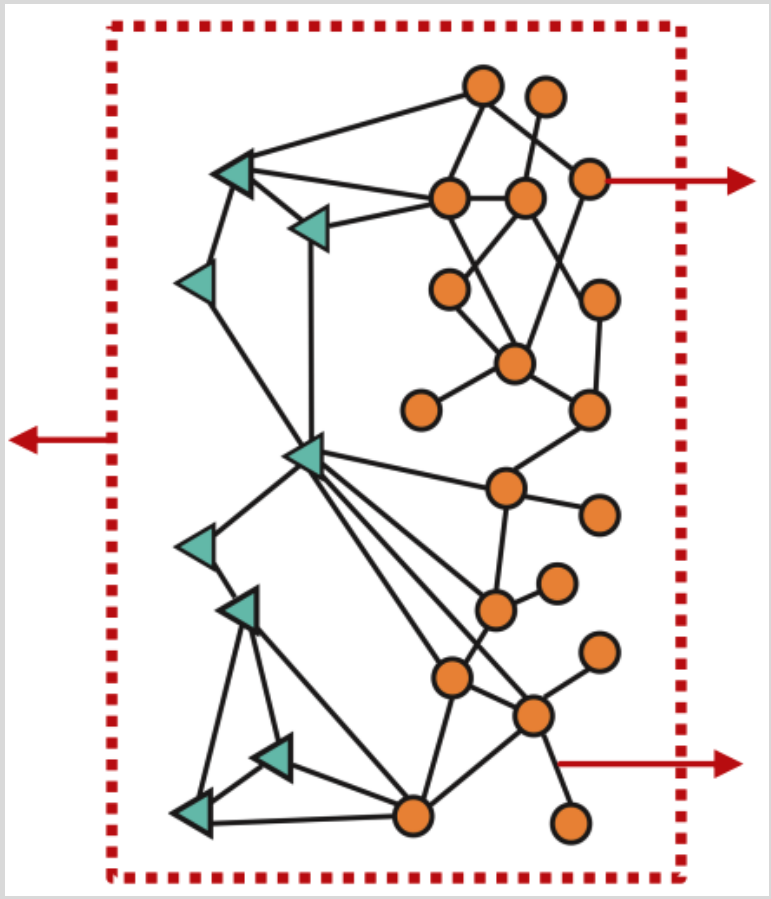


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Node-level prediction head

- Use the node embedding as input to a prediction function (e.g., an MLP)
- Following the GNN computation with K layers, we have a d_K-dimensional embeddings: $\left\{\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}, \forall v \in V\right\}$
- Assume we want to make a C-way prediction
  - Classification: classify among C categories
  - Regression: regress on C variables
- Create a prediction head, g, that outputs and estimate, $\hat{\boldsymbol{y}}_v$, of the target, from $\mathbf{h}_v^{(K)}$
  - Example: g is a linear layer, $\hat{\boldsymbol{y}}_v = g\left(\mathbf{h}_v^{(K)}\right) = W^g \mathbf{h}_v^{(K)}$
    - $W^g \in \mathbb{R}^{C \times d_K} \rightarrow$ maps node embeddings from $\mathbb{R}^{d_K}$ to $\mathbb{R}^C$
    - We can now compute a loss function on the node prediction task!

Edge-level prediction head

- Use pairs of node embeddings as input to a prediction function
- Assume we want to make a C-way prediction
- $\hat{\mathbf{y}}_{uv} = g\left(\mathbf{h}_v^{(K)}, \mathbf{h}_u^{(K)}\right)$
- What are the options for $g\left(\mathbf{h}_v^{(K)}, \mathbf{h}_u^{(K)}\right)$?

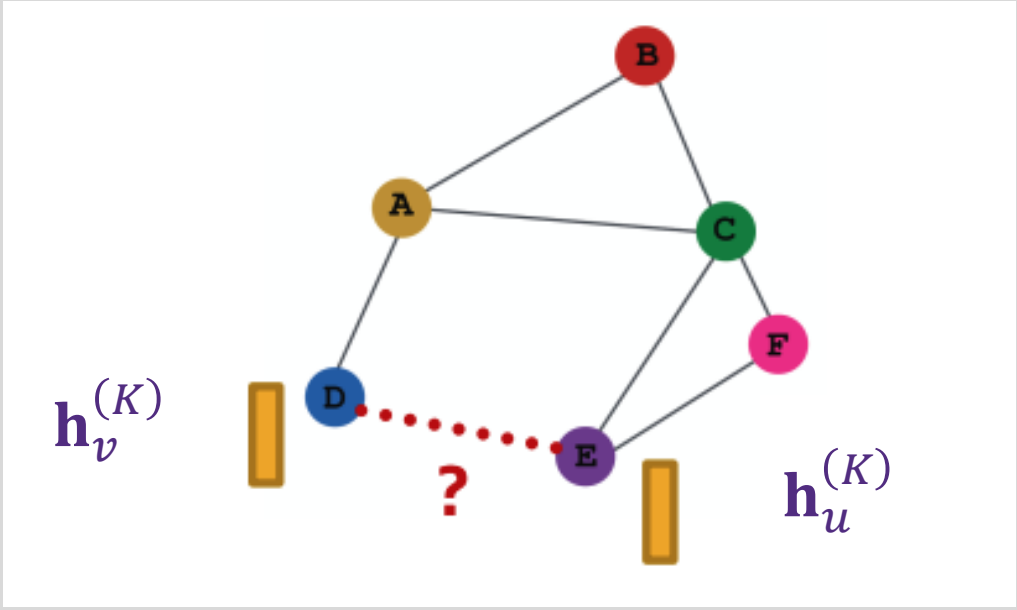
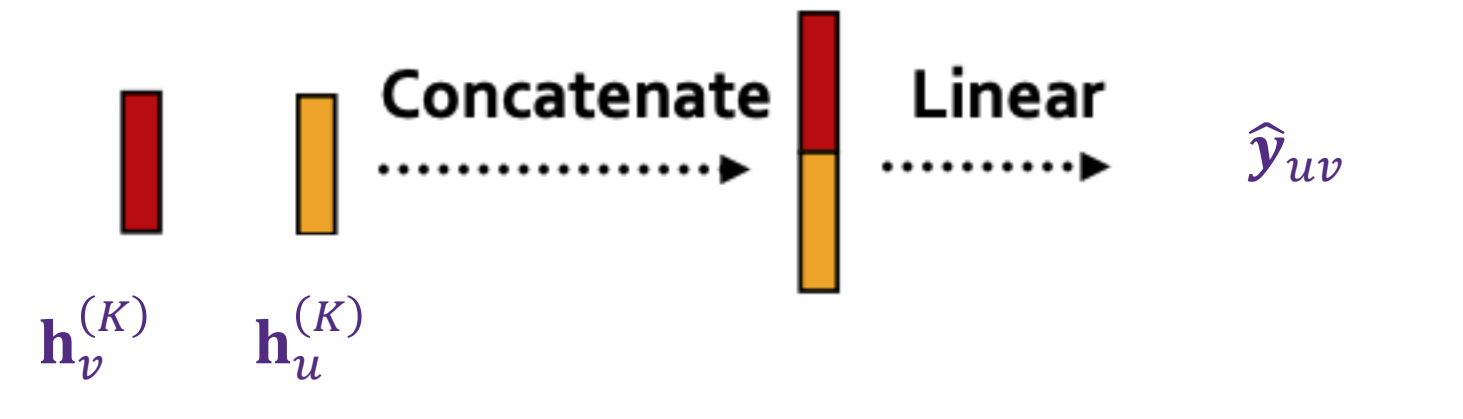


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Edge-level prediction head with concatenation

- One option is to concatenate the u and v node embeddings
- Prediction function, g, operates on the concatenated vector
- Example: g is a linear layer, $\hat{y}_{uv} = g\left(\left[\mathbf{h}_u^{(K)} \mid \mathbf{h}_v^{(K)}\right]\right) = W^g\left[\mathbf{h}_u^{(K)} \mid \mathbf{h}_v^{(K)}\right]$



- Here, $W^g \in \mathbb{R}^{C \times 2d_K} \rightarrow$ maps node embeddings from $\mathbb{R}^{2d_K}$ to $\mathbb{R}^C$

Account for the concatenation of embeddings

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Edge-level prediction head with dot product

- The prediction head function is just the dot product of the node embeddings
  - $\hat{\boldsymbol{y}}_{uv} = \mathbf{h}_u^{(K)^T} \cdot \mathbf{h}_v^{(K)}$
  - This only applies to 1-way prediction (i.e., binary classification, single regression, link prediction)
- Applying to C-way prediction with a learnable generalized dot product
  - Learn c weight matrices

$$\hat{y}_{uv,1} = \mathbf{h}_u^{(K)^T} \mathbf{W}^1 \mathbf{h}_v^{(K)}$$
$$\vdots$$
$$\hat{y}_{uv,C} = \mathbf{h}_u^{(K)^T} \mathbf{W}^C \mathbf{h}_v^{(K)}$$
$$\hat{\boldsymbol{y}}_{uv} = \left[\hat{y}_{uv,1}, \cdots, \hat{y}_{uv,C}\right] \in \mathbb{R}^C$$

where $W^c \in \mathbb{R}^{d_K \times d_K}$

Graph-level prediction head

- Assume we want to make a C-way prediction
- $\hat{\boldsymbol{y}}_G = g\left(\left\{\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}, \forall v \in V\right\}\right)$
- Options for g are similar to message aggregation and update

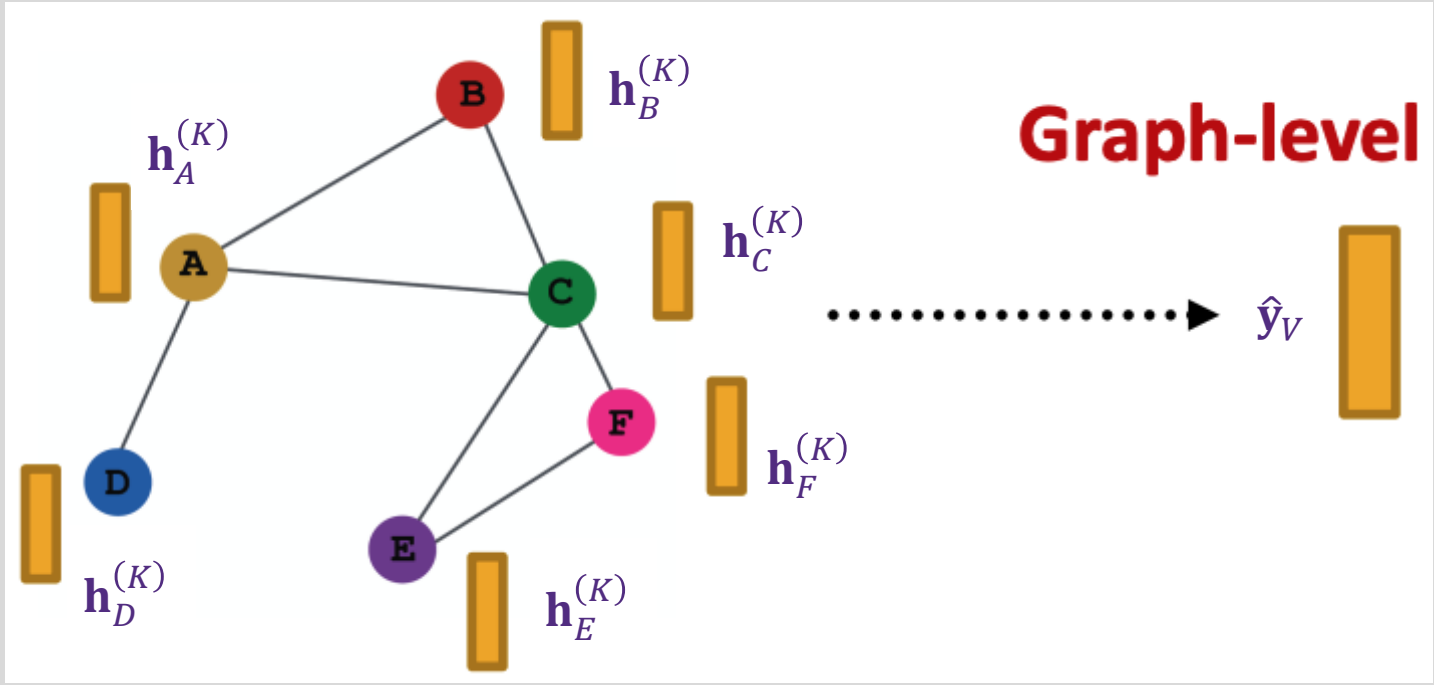
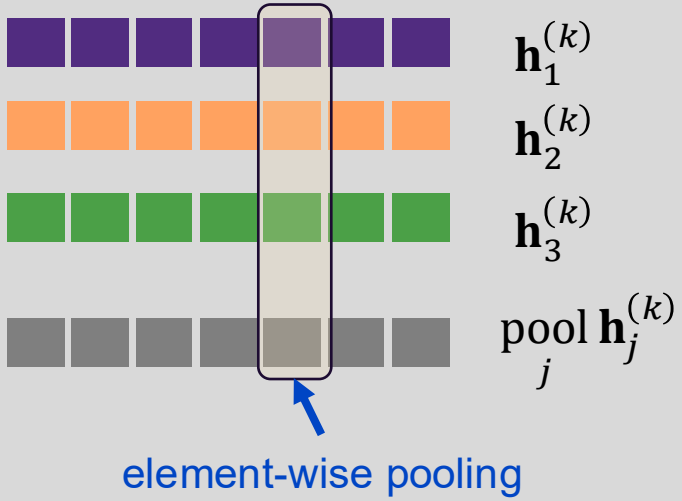


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Graph-level prediction head via pooling

- Element-wise pooling (aggregation) over node embeddings
  - Mean pooling: $\mathbf{h}_G^{(K)} = Mean\left(\left\{\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}, \forall v \in V\right\}\right)$
  - Max pooling: $\mathbf{h}_G^{(K)} = Max\left(\left\{\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}, \forall v \in V\right\}\right)$
  - Sum pooling: $\mathbf{h}_G^{(K)} = Sum\left(\left\{\mathbf{h}_v^{(K)} \in \mathbb{R}^{d_K}, \forall v \in V\right\}\right)$



- Prediction head operates on pooled embedding

$$\hat{\boldsymbol{y}}_G = g\left(\mathbf{h}_G^{(K)}\right)$$

  - Example: g is a linear layer, $\hat{\boldsymbol{y}}_G = g\left(\mathbf{h}_G^{(K)}\right) = W^g \mathbf{h}_G^{(K)}$
    - $W^g \in \mathbb{R}^{C \times d_K} \rightarrow$ maps node embeddings from $\mathbb{R}^{d_K}$ to $\mathbb{R}^C$
- Works well for small graphs. What about large graphs?

Limitations for global pooling

- Global pooling loses information
- Toy example with 1-dimensional node embeddings for two graphs each with 5 nodes
  - Node embeddings for G_1: $\{-1, -2, 0, 1, 2\}$
  - Node embeddings for G_2: $\{-10, -20, 0, 10, 20\}$
  - G_1 and G_2 have different embeddings ⟶ we should be able to differentiate them with appropriate prediction head
- But, if we do global sum pooling:
  - $\hat{\boldsymbol{y}}_{G_1} = g\left(\mathbf{h}_{G_1}^{(K)}\right) = g(sum\{-1, -2, 0, 1, 2\}) = g(0)$
  - $\hat{\boldsymbol{y}}_{G_2} = g\left(\mathbf{h}_{G_2}^{(K)}\right) = g(sum\{-10, -20, 0, 10, 20\}) = g(0)$
  - We cannot differentiate G_1 and G_2

Hierarchical global pooling

- Aggregate the node embeddings hierarchically
- Toy example from previous slide with hierarchical aggregation with ReLU
  - Separately aggregate first 2 nodes and last 3 nodes
  - Aggregate again for final prediction
- Node embeddings for $G_1: \{-1, -2, 0, 1, 2\}$
  - Round 1: $h_{G_1,a} = ReLU\big(sum(-1,-2)\big) = 0$, $h_{G_1,b} = ReLU\big(sum(0,1,2)\big) = 3$
  - Round 2: $h_{G_1} = ReLU\left(sum\big(h_{G_1,a}, h_{G_1,b}\big)\right) = 3$
- Node embeddings for $G_2: \{-10, -20, 0, 10, 20\}$
  - Round 1: $h_{G_2,a} = ReLU\big(sum(-10,-20)\big) = 0$, $h_{G_2,b} = ReLU\big(sum(0,10,20)\big) = 30$
  - Round 2: $h_{G_2} = ReLU\left(sum\big(h_{G_2,a}, h_{G_2,b}\big)\right) = 30$
- $\hat{\boldsymbol{y}}_{G_1} = g(3), \hat{\boldsymbol{y}}_{G_2} = g(30)$. We can differentiate these!

Differentiable pooling

- Idea: learn a hierarchical node pooling

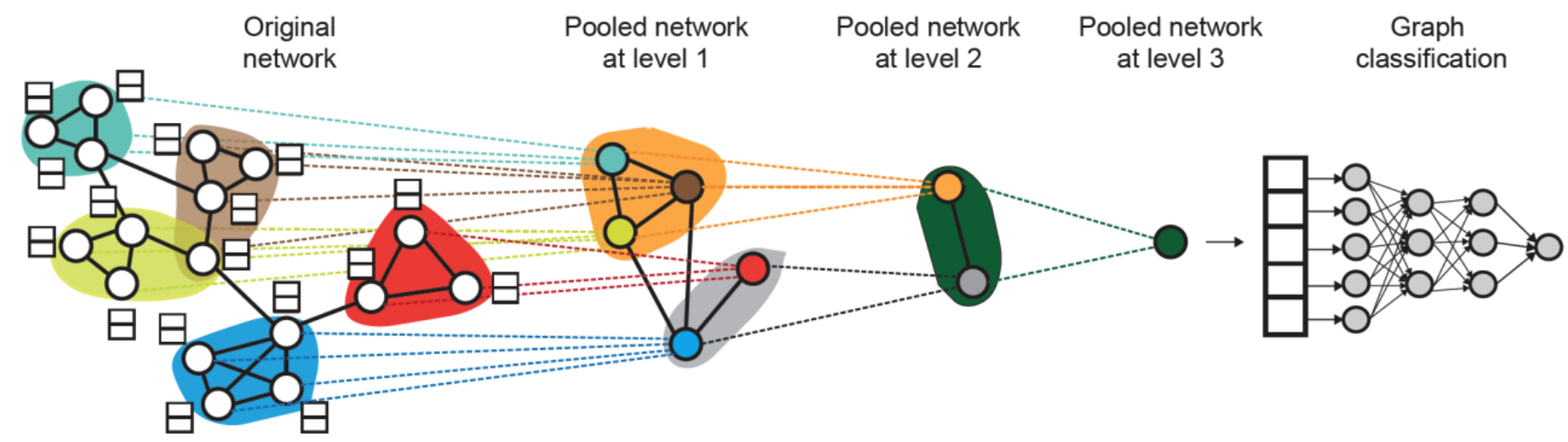


Image credit: Ying et. al. 2019

- Leverage 2 independent GNNs at each pooling level
  - GNN 1 computes node embeddings
  - GNN 2 computes node cluster membership
- GNN 1 and 2 at each level can be executed in parallel

Differentiable pooling

- Idea: learn a hierarchical node pooling

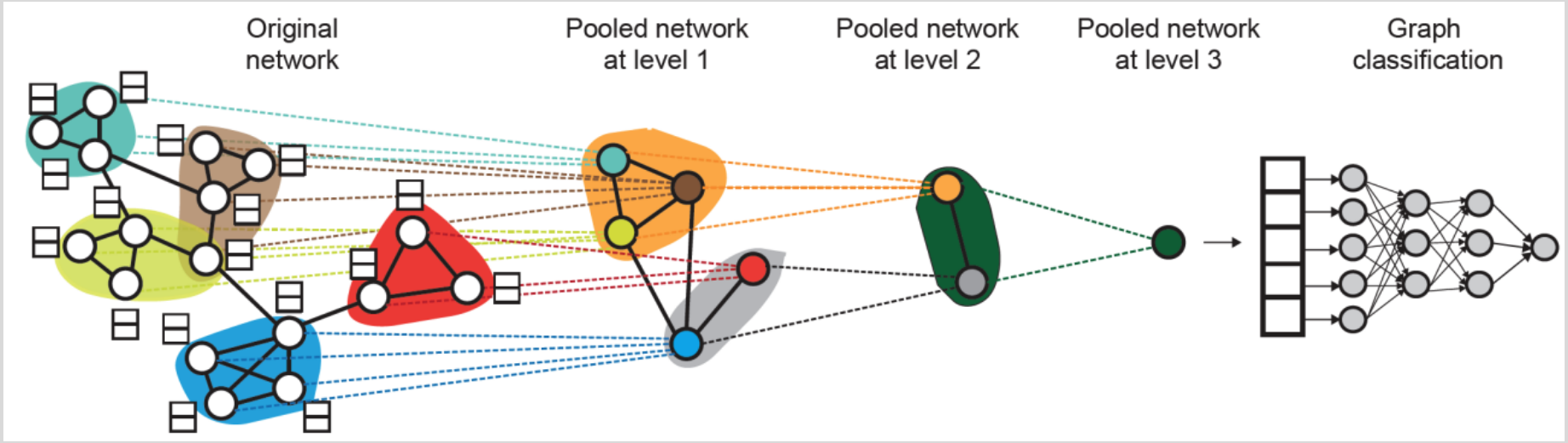


Image credit: Ying et. al. 2019

- For each pooling layer
  - Use cluster assignments to aggregate node embeddings
  - Create a single new node for each cluster, maintaining edges between clusters to generate next level pooled network
- Jointly train GNN 1 and GNN 2

GNN training

Where do ground-truth labels come from?

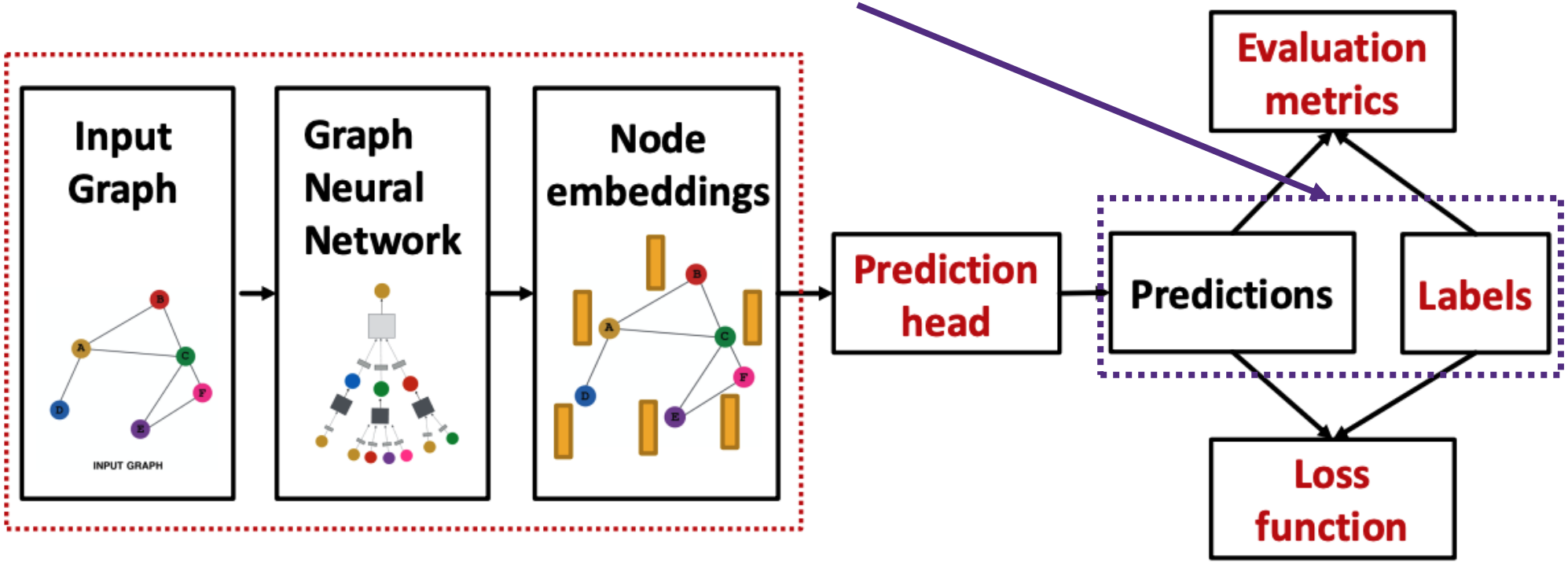


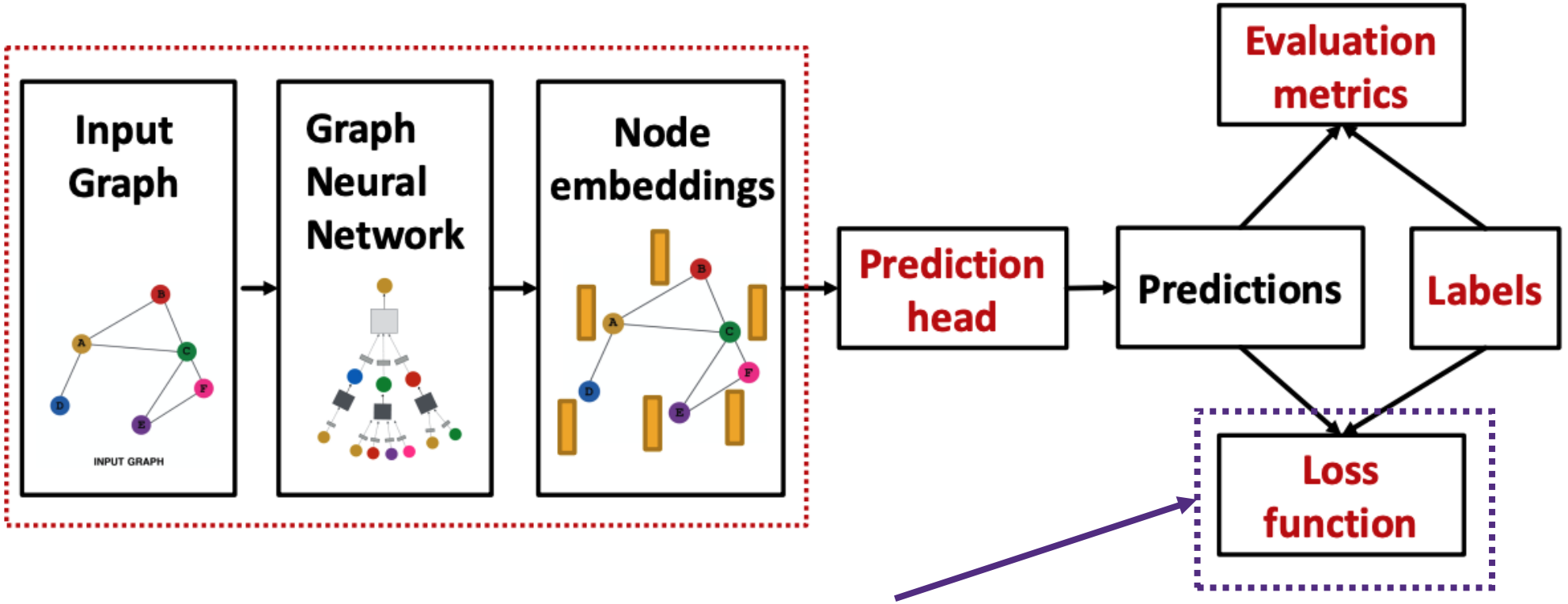
Image credit: Jure Leskovec, http://cs224w.stanford.edu

Supervised learning

- Supervised learning – labels come from an external source
- Examples
  - Node labels: article topic in a citation network
  - Edge labels: fraudulent transaction in a financial transaction network
  - Graph labels: toxicity of a drug

Unsupervised learning

- Unsupervised learning – labels come from the graph structure
- Examples
  - Node labels: node statistics such as a centrality measure
  - Edge labels: link prediction based on input graph structure
  - Graph labels: graph statistics such as diameter, isomorphism (pair of graphs)
- Technically, we have supervision in unsupervised
- Unsupervised is more appropriately called self-supervised



How do we compute the final loss?

Image credit: Jure Leskovec, http://cs224w.stanford.edu

GNN training setting

- We have N data points
  - Each data point, i, can be a node/edge/graph
  - Node-level: prediction $\hat{\boldsymbol{y}}_v^{(i)}$, label $\boldsymbol{y}_v^{(i)}$
  - Edge-level: prediction $\hat{\boldsymbol{y}}_{uv}^{(i)}$, label $\boldsymbol{y}_{uv}^{(i)}$
  - Graph-level: prediction $\hat{\boldsymbol{y}}_G^{(i)}$, label $\boldsymbol{y}_G^{(i)}$
- To simplify notation, we'll drop the subscripts and use predictions $\hat{\boldsymbol{y}}^{(i)}$ and label $\boldsymbol{y}^{(i)}$ for all task data types

Classification or regression

- Classification: labels $\boldsymbol{y}^{(i)}$ are categorical values (no objective ordering)
- Regression: labels $\boldsymbol{y}^{(i)}$ are numerical (typically continuous, ordered)
- GNNs are applicable in both settings but with different loss functions and evaluation metrics

Classification loss functions

- Cross entropy (CE) is the most common
- Binary CE (two classes) for the i^{th} data point

$$BCE\left(y^{(i)}, \hat{y}^{(i)}\right) = y^{(i)} log \hat{y}^{(i)} + \left(1 - y^{(i)}\right) log\left(1 - \hat{y}^{(i)}\right)$$

- C-way CE (C classes) for the i^{th} data point

$$CE\left(\boldsymbol{y}^{(i)}, \hat{\boldsymbol{y}}^{(i)}\right) = -\sum_{j=1}^{C} \boldsymbol{y}_j^{(i)} log \hat{\boldsymbol{y}}_j^{(i)} = -\boldsymbol{y}^{(i)} \cdot \hat{\boldsymbol{y}}^{(i)}$$

  where $\boldsymbol{y}^{(i)} \in \mathbb{R}^C$ is a one-hot label encoding, $\hat{\boldsymbol{y}}^{(i)} \in \mathbb{R}^C$ is prediction after Softmax

- Total loss over all N training samples

$$\mathcal{L} = \sum_{i=1}^{N} B/CE\left(\boldsymbol{y}^{(i)}, \hat{\boldsymbol{y}}^{(i)}\right)$$

- There are many other loss functions – we'll cover them as needed

Regression loss function

- Most often use Mean Squared Error (MSE)
- C-way MSE (C outcome variables) for the i^{th} data point

$$MSE\left(\boldsymbol{y}^{(i)}, \hat{\boldsymbol{y}}^{(i)}\right) = -\sum_{j=1}^{C}\left(\boldsymbol{y}_j^{(i)} - \hat{\boldsymbol{y}}_j^{(i)}\right)^2$$

where $\boldsymbol{y}^{(i)} \in \mathbb{R}^C$ is a real valued vector of target outcomes, $\hat{\boldsymbol{y}}^{(i)} \in \mathbb{R}^C$ is a real valued vector of predicted outcomes

- Total loss over all N training samples

$$\mathcal{L} = \sum_{i=1}^{N} MSE\left(\boldsymbol{y}^{(i)}, \hat{\boldsymbol{y}}^{(i)}\right)$$

GNN performance evaluation

How do we measure performance?

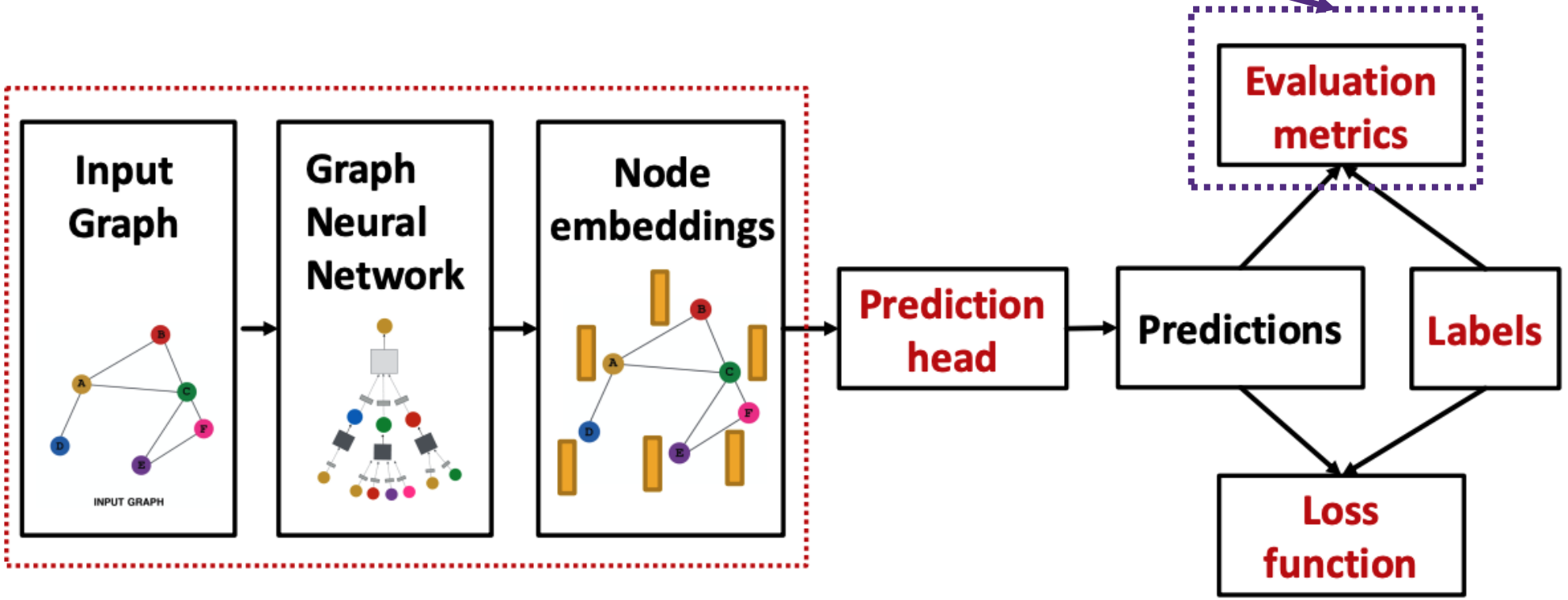


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Evaluation metrics for regression

- Use standard machine learning evaluation metrics
- Assume we make predictions for M data points
- Root mean square error (RMSE)

$$\sqrt{\frac{1}{M}\sum_{i=1}^{M}(\boldsymbol{y}^{(i)} - \hat{\boldsymbol{y}}^{(i)})^2}$$

- Mean absolute error

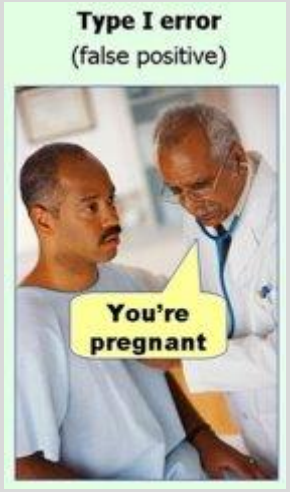
$$\frac{1}{N}\left|\boldsymbol{y}^{(i)} - \hat{\boldsymbol{y}}^{(i)}\right|$$

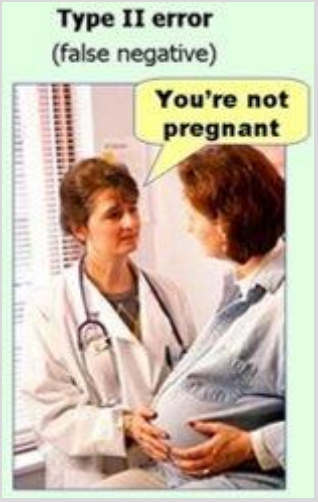
- Note, for C-way regression, these result in a vector – one entry for each target outcome

Classifier error types

All classifier performance metrics are derived from counts of TP, FP, TN, FN

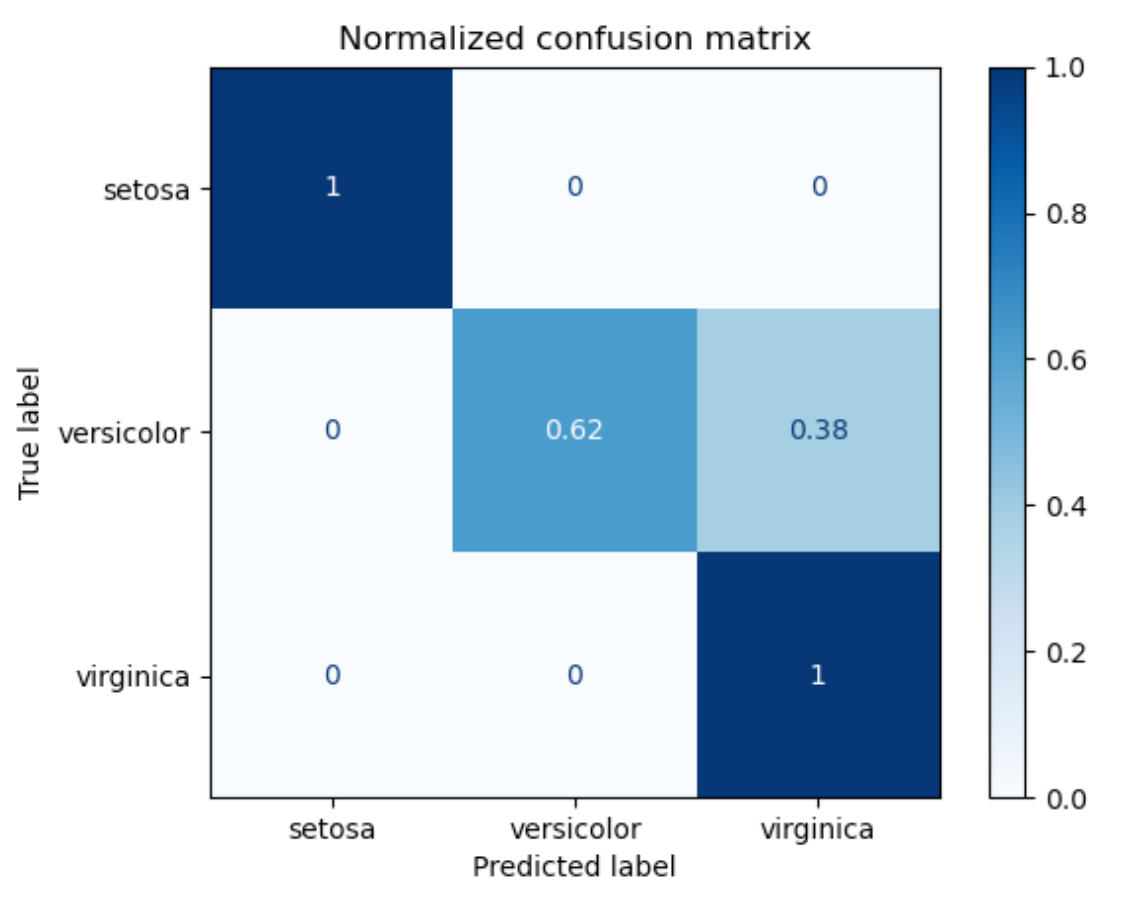
		Actual Class: Positive	Actual Class: Negative
Predicted Class	Positive	True Positive (TP)	False Positive (FP) *False Alarm* *Type I Error*
Predicted Class	Negative	False Negative (FN) *Type II Error*	True Negative (TN)





Confusion Matrix

We can identify how correctly classified were the observations by computing a Confusion Matrix that shows the correct and misclassified observations.

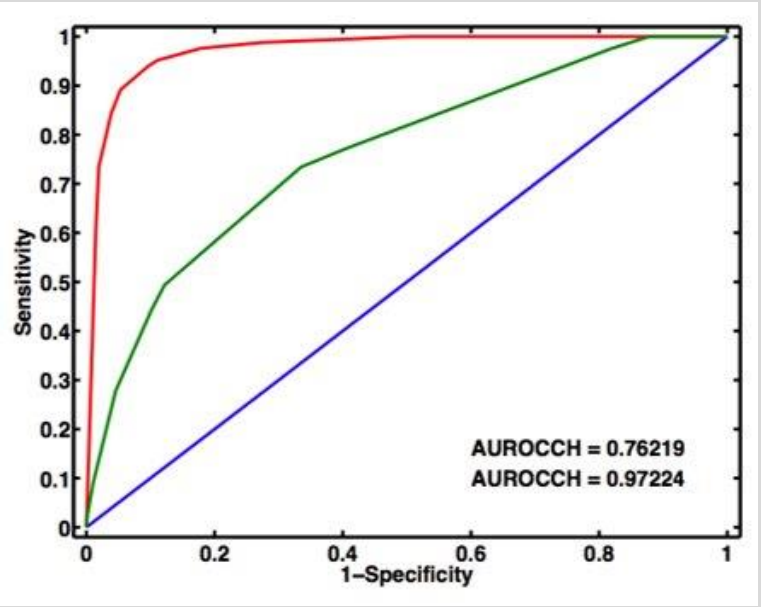


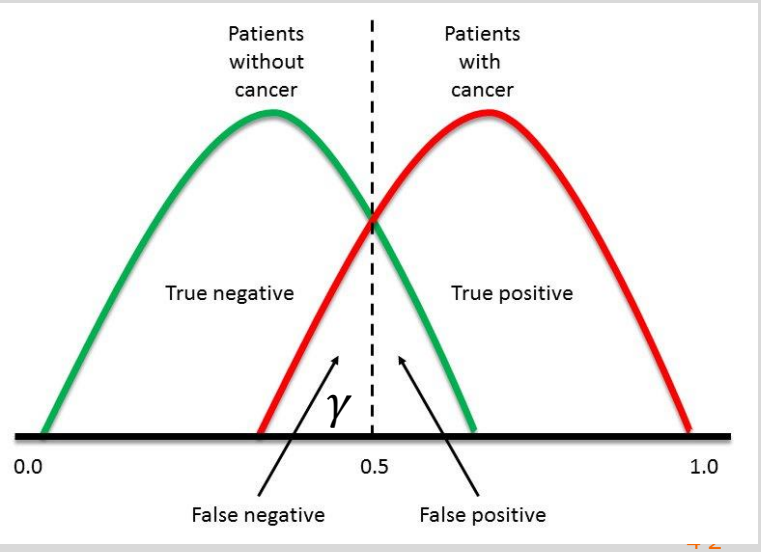
Classifier performance point metrics

- Let P and N be the number of positive and negative samples, respectively
- Accuracy = (TP+TN)/(TP+TN+FP+FN)
- Sensitivity (a.k.a Recall, True Positive Rate) = TP/(TP+FN) = TP/P
  - The fraction of actually positive samples predicted positive by the model
- Specificity (a.k.a. True Negative Rate) = TN/(TN+FP) = TN/N
  - The fraction of actually negative samples predicted negative by the model
- Precision (a.k.a. Positive Predictive Value) = TP/(TP+FP)
  - The fraction of samples predicted positive by the model that are actually positive
- False Positive Rate (a.k.a. Probability of false alarm) = FP/(FP+TN) = FP/N
- F1 Score = 2(Precision*Recall)/(Precision + Recall)
- Balanced Accuracy = (Sensitivity + Specificity)/2

Classifier performance with ROC and AUC

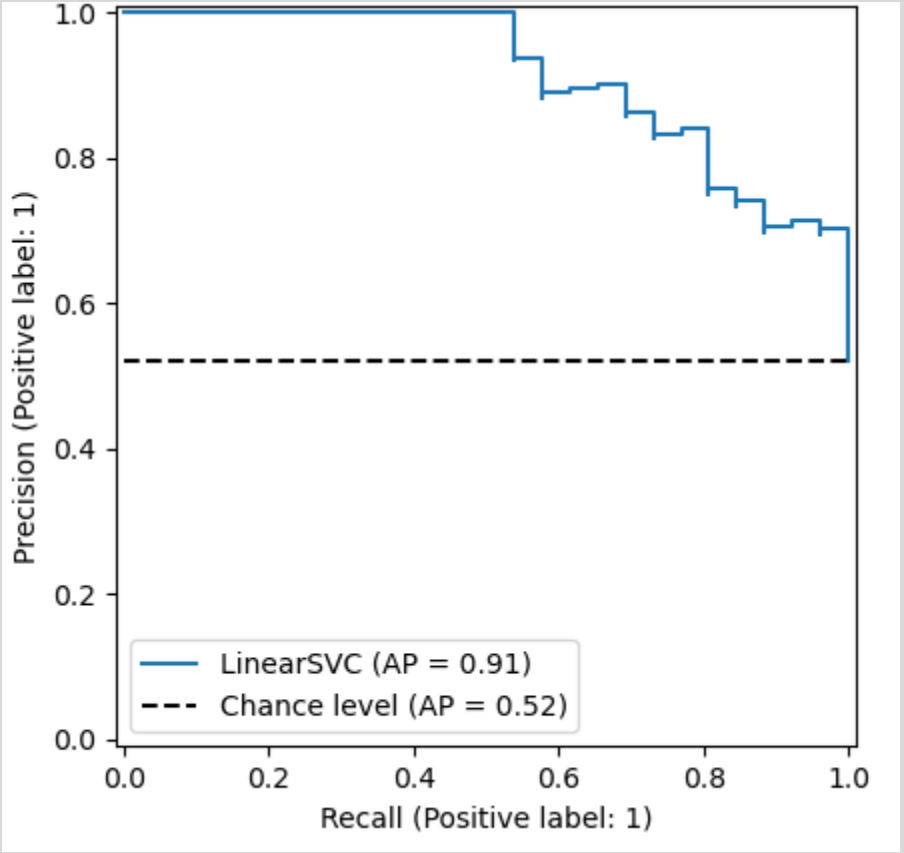
- Recall, we can select $0 < \gamma < 1$ as a threshold s.t. $\hat{P}(Y = 1) \geq \gamma$ implies $\hat{y} = 1$. What happens if we change γ?
- Receiver Operating Characteristic Curve (ROC)
  - Illustrates tradeoff between FPR and TPR of a binary classifier as the discrimination threshold, γ, is adjusted
  - Axes
    - 1 – Specificity (false positive rate) = 1 – TN/N
    - Sensitivity (true positive rate) = TP/P
- Area under the curve (AUC)
  - Summary statistic
  - AUC = 0.5 indicates no predictive value (random guessing)
  - AUC = 1.0 indicates perfect predictive value





Classifier performance with precision-recall curve

- Illustrates tradeoff between precision (PPV) and sensitivity of a binary classifier as the discrimination threshold, γ, is adjusted
- Particularly useful for highly imbalanced datasets (i.e., where one class is very rare). In those cases, it's possible to achieve high AUC with very low precision*
- Axes
  - Sensitivity (true positive rate) = TP/P
  - Precision (PPV) = TP/(TP+FP)
- Average precision – summary metric indicating the weighted average precision for each discrimination threshold



*Romero-Brufau S, Huddleston JM, Escobar GJ, Liebow M. Why the C-statistic is not informative to evaluate early warning scores and what metrics to use. Crit Care. 2015 Aug 13;19(1):285

Splitting our dataset

Dataset splits

- Assume we're working with a large dataset and not using cross-validation
- Randomly split the data into:
  - Training set: used for optimizing GNN and prediction head parameters
  - Validation set: used to develop model (different architectures) and tune hyperparameters
  - Test set: held out until model(s) are finalized to report final performance
- If feasible, we should repeat the process and report average performance over many different random splits
- Concern: Can we guarantee that the test set is really held out?

Graph data splitting is special

- Suppose we are splitting an image dataset for image classification
- Each data point is an image
- Data points are independent
  - Image 5 will not affect our prediction for image 1

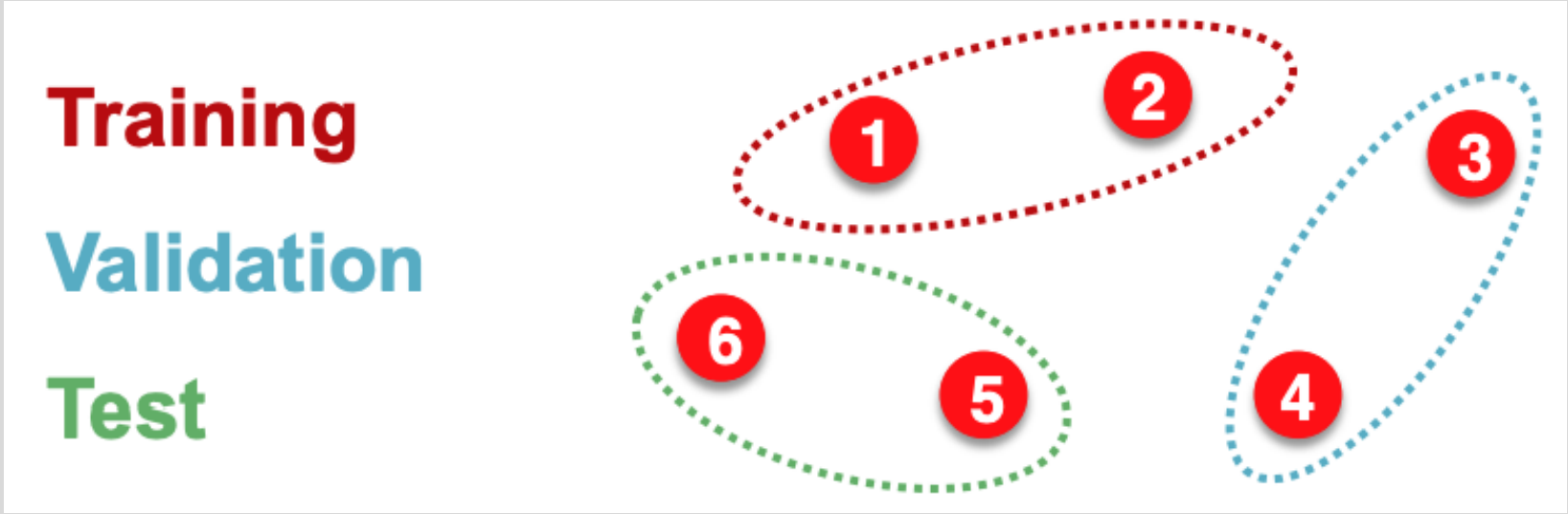
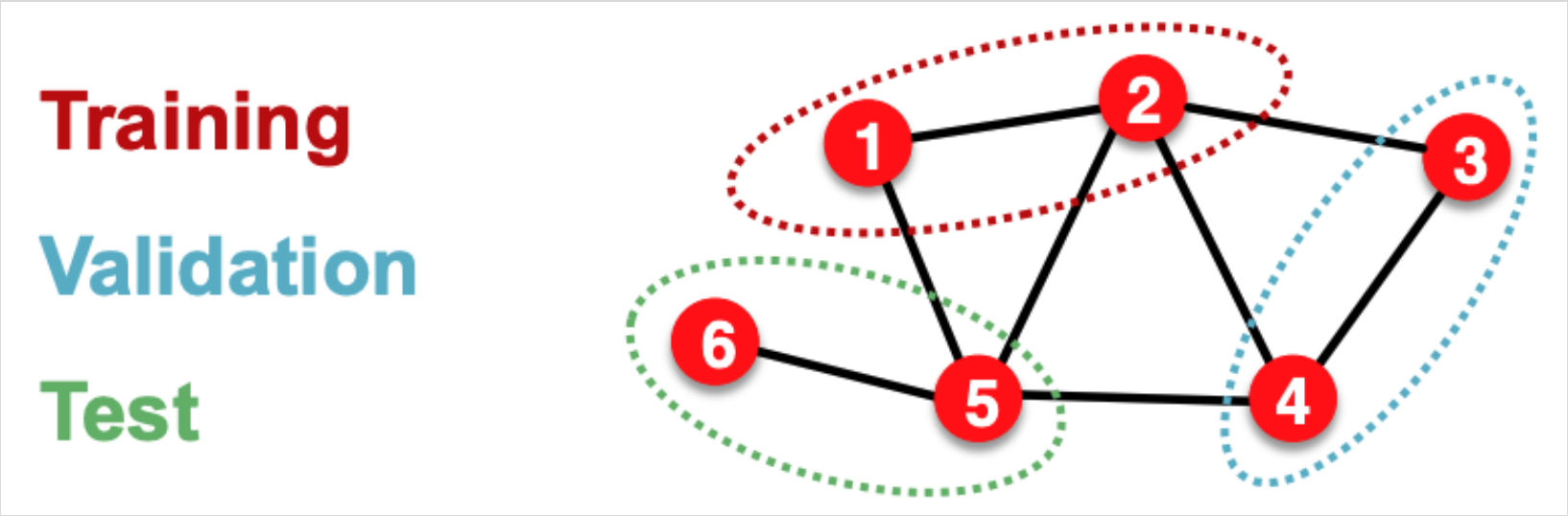


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Graph data splitting is special

- Assume we wish to perform node classification
- Each data point is a node
- Data points are NOT independent
  - Node 5 will affect our prediction for node 1 because it will participate in message passing that will affect node 1's embedding



Transductive setting data splits

- The input graph is observed in all data splits (train, validation, test)
- Only the node labels are split
  - Training time, compute embeddings using the entire graph via message passing, evaluate the prediction head performance and update the model parameters using training nodes (1 & 2 in example)
  - Validation time, compute embeddings using the entire graph and evaluate the prediction head performance on validation nodes (3 & 4 in example)

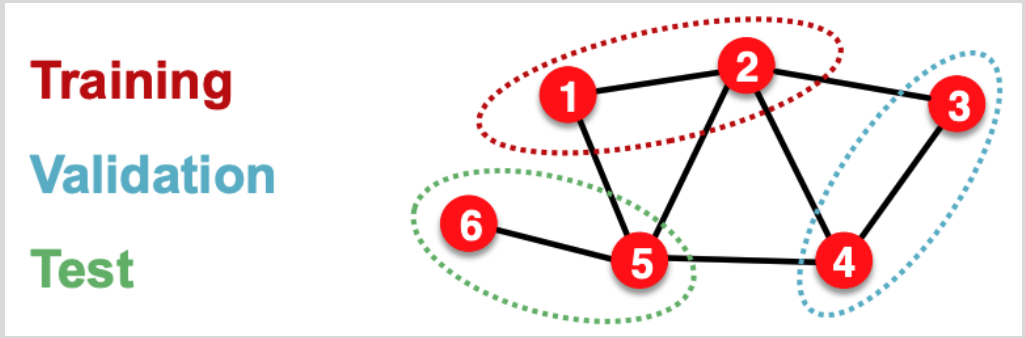


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Inductive setting data splits

- Break edges between the splits (generates multiple graphs / components)
  - Now we have three independent graphs
  - Training time, compute embeddings using only the training graph, evaluate the prediction head performance and update model parameters using training nodes (1 & 2 in example)
  - Validation time, compute embeddings for the validation nodes and evaluate performance on validation nodes (3 & 4 in example)

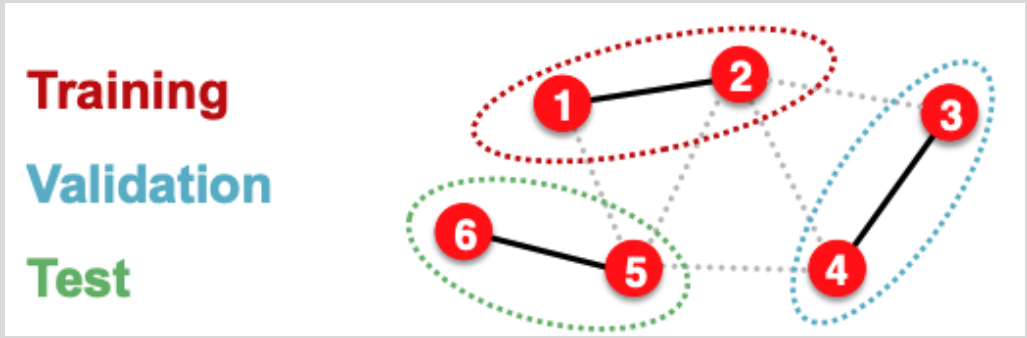
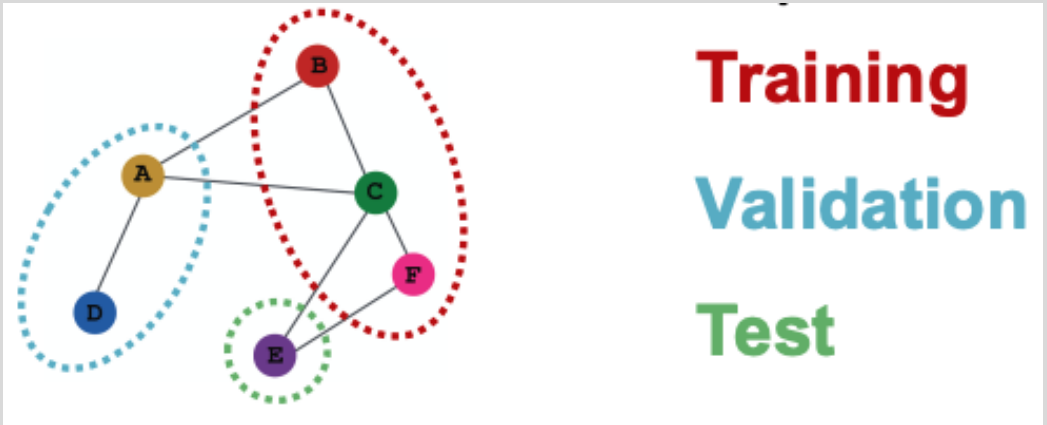


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Node classification

- Transductive – all splits can observe the entire graph structure, but can only observe the labels of their respective nodes



- Inductive – each split contains an independent graph

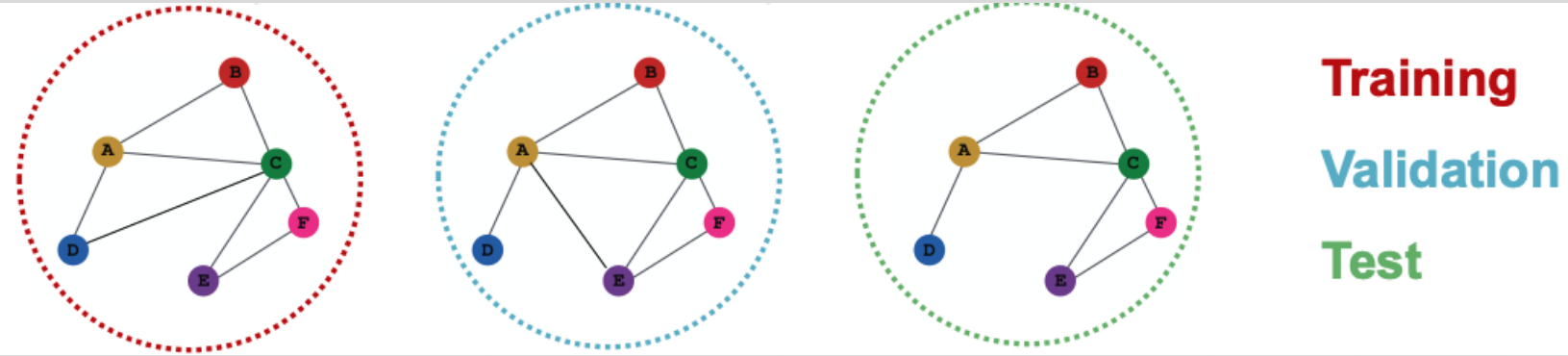


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Graph classification

- Only the inductive setting is applicable because we must test on unseen graphs

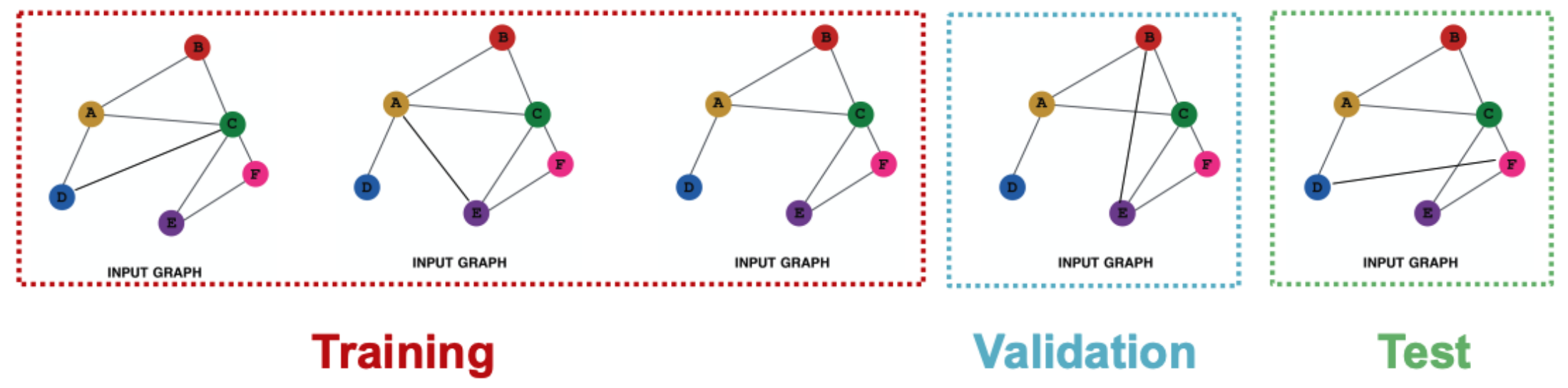


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Edge (link) prediction

- Goal is to predict missing edges
- We need to hide some edges in the input graph from the GNN and let the GNN + prediction head determine if the edge should exist

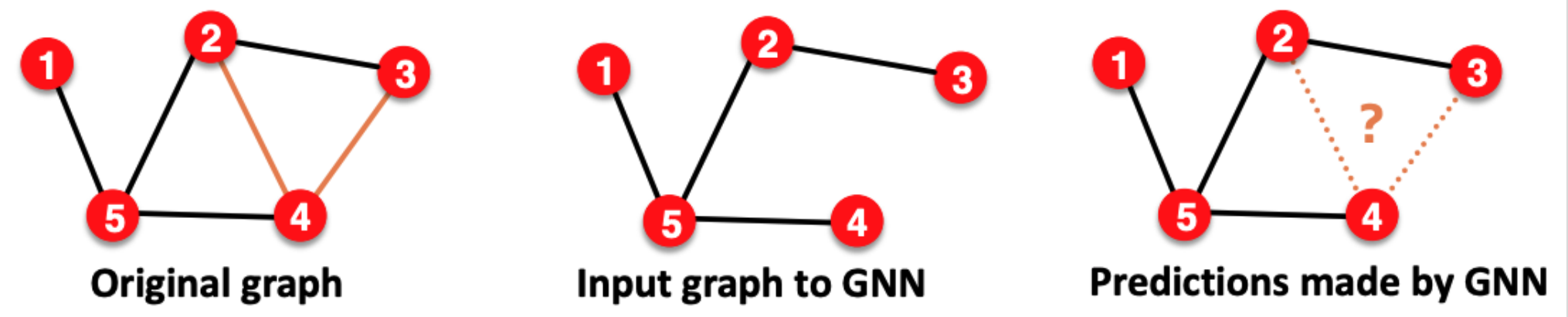


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Setting up edge prediction

- We will need to split the edges twice
- First split: assign 2 types of edges in the input graph
  - Message edges: will be used for GNN message passing
  - Supervision edges: will be used for training and evaluation
- After the first split:
  - Only message edges will remain in the input graph
  - Supervision edges are not fed into the GNN

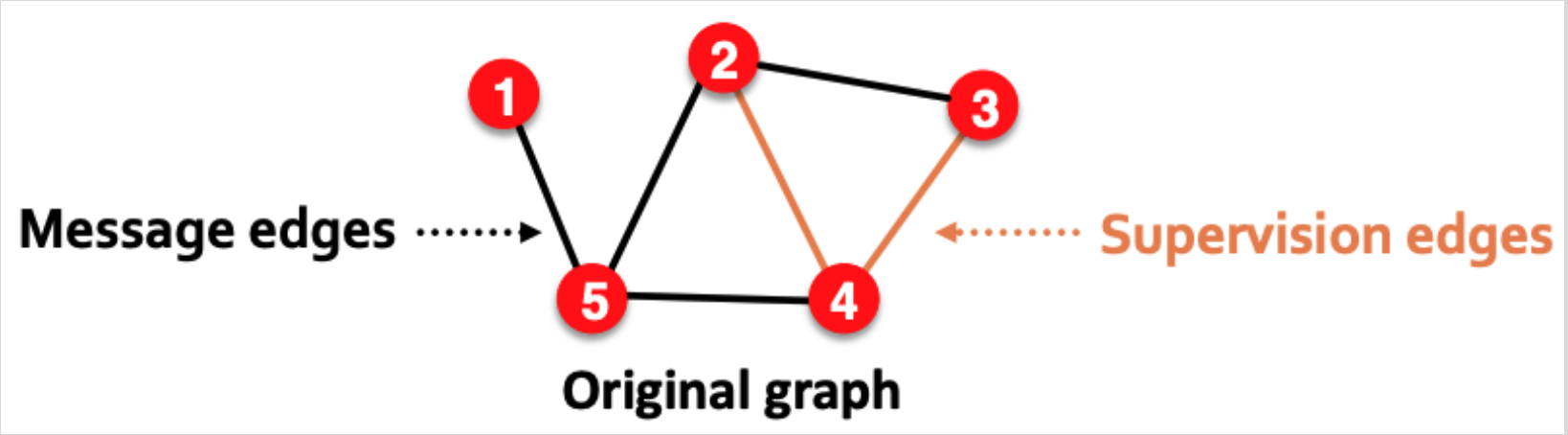


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Setting up edge prediction

- Option 1: Inductive second split: assign each message & supervision edge to one of train / validation / test
- The assigned edges will form three independent graphs
- Each graph will have message and supervision edges

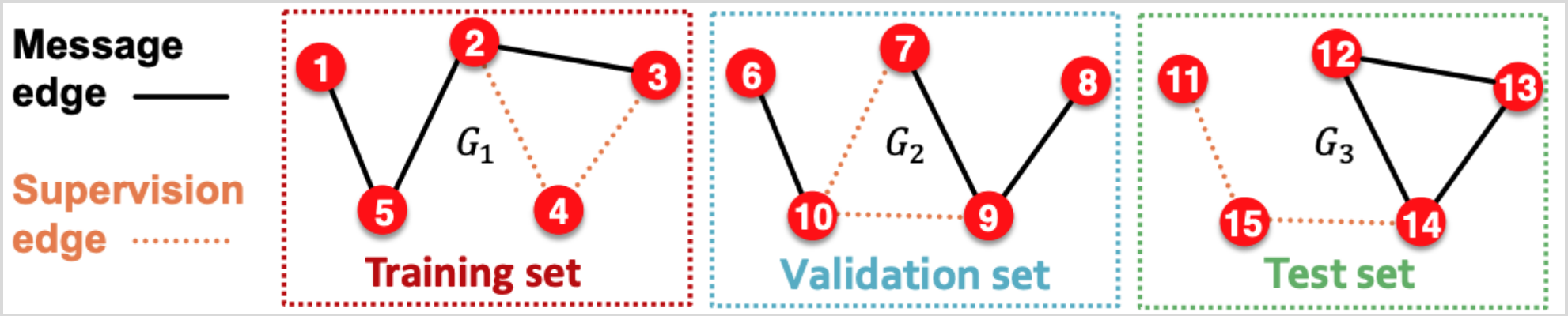


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Setting up edge prediction

- Option 2: Transductive second split (most common choice)
  - All message edges used in training, supervision edges split among train / val / test

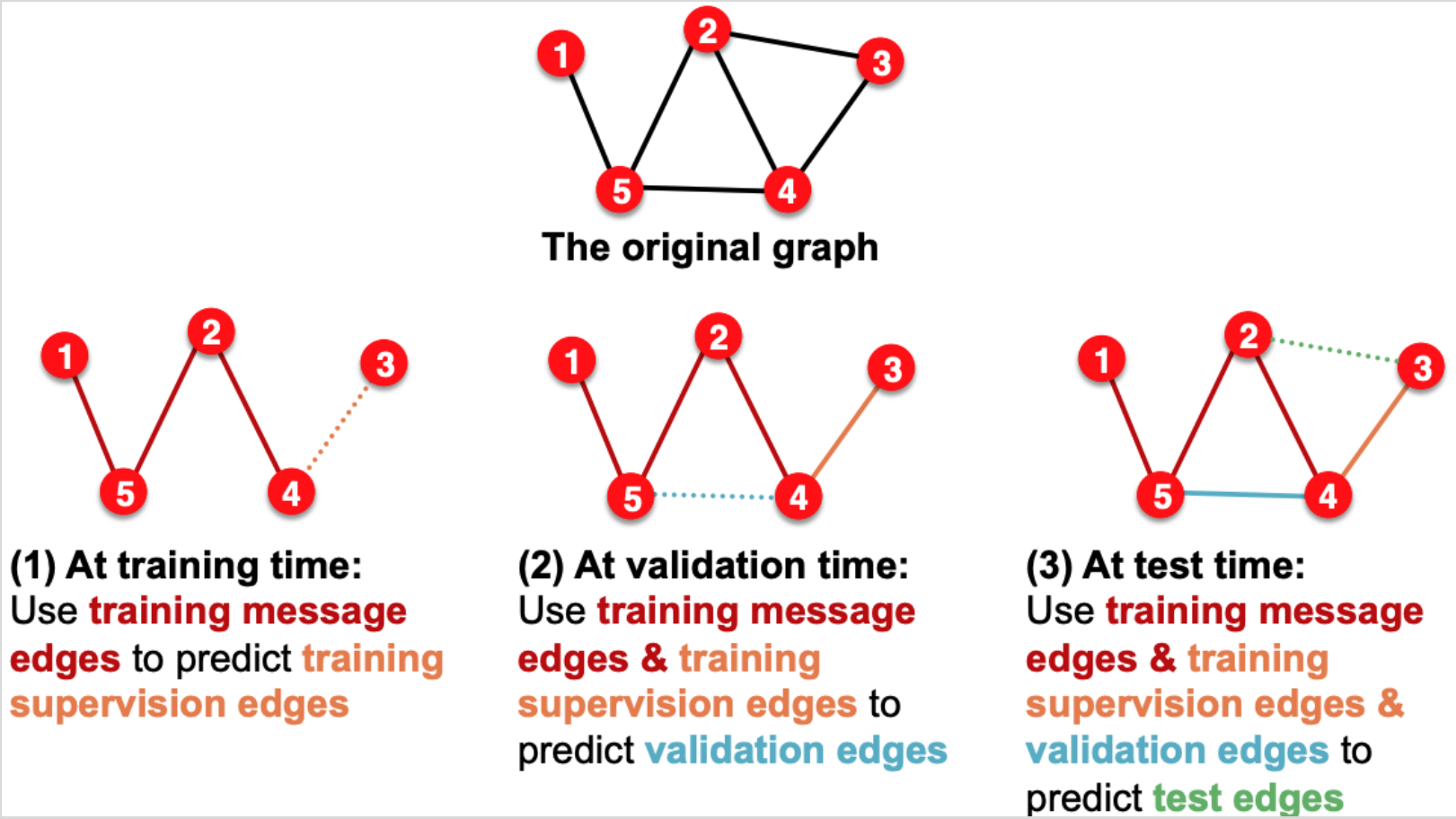
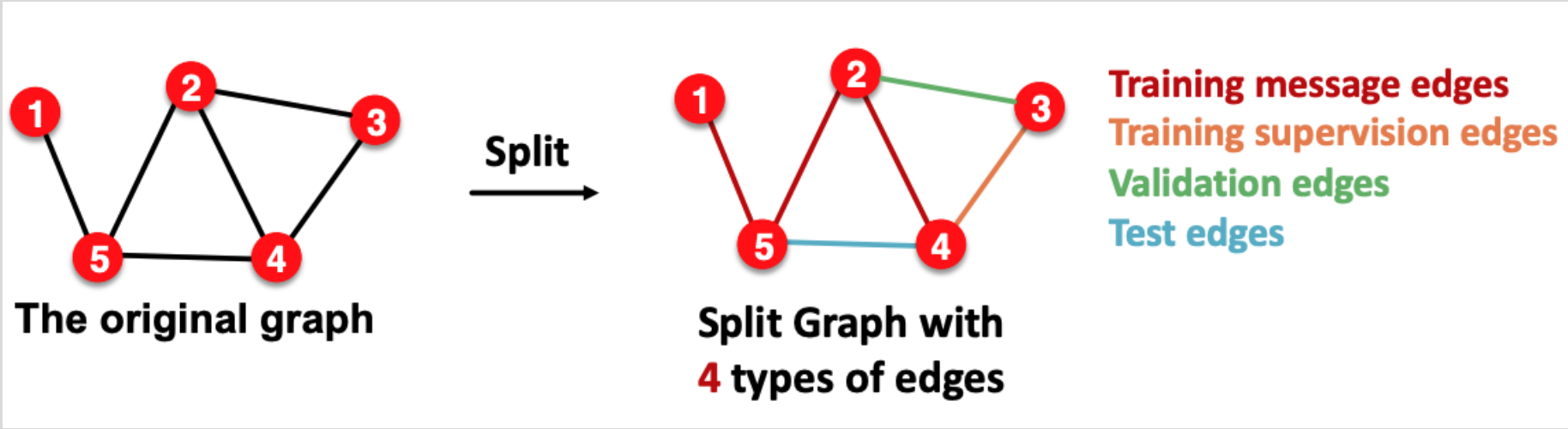


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Setting up edge prediction

- Option 2: Transductive second split (most common choice)
- Link prediction settings are tricky. You may see variations in papers.



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